A CRITERION FOR A MAP OF MODULES TO BE AN ISOMORPHISM IN TERMS OF DEPTH AND ASSOCIATED PRIMES

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Allowed standard prerequisites (editorial).

- Injectivity criterion on associated primes. Lemma \ref{lemma-check-injective-on-ass}, cited at proof lines 5466-5467. Gives injectivity of the map under the stated hypotheses.
- The associated primes of a localisation and the depth of a localisation. Algebra, Lemma \ref{algebra-lemma-ideal-nonzerodivisor}, cited at proof lines 5477-5479. Turns the condition that a prime is not associated into a lower bound on the depth of the target.
- Behaviour of depth in a short exact sequence. Algebra, Lemma \ref{algebra-lemma-depth-in-ses}, cited at proof lines 5483-5485. Shows the cokernel has depth at least one at every prime where it need not already vanish.
- A module with no associated primes is zero. Algebra, Lemma \ref{algebra-lemma-ass-zero}, cited at proof lines 5485-5486. Concludes that the cokernel vanishes.

Source reference. Original lemma 0AV8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13676–13676. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AV7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13675–13675. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00LL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 791–791. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 803–803. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0587: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6770–6770. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Let $\varphi : M \to N$ be a map of R-modules. Assume M is finite and that for every prime $\mathfrak{p}$ of R one of the following happens*

(1) *$M_\mathfrak{p} \to N_\mathfrak{p}$ is an isomorphism, or*
(2) *$\mathrm{depth}(M_\mathfrak{p}) \geq 2$ and $\mathfrak{p} \notin \mathrm{Ass}(N)$.*

Then φ is an isomorphism.

Proof. By Lemma 0AV7 we see that φ is injective. Let $N' \subset N$ be an finitely generated R-module containing the image of M. Then $\mathrm{Ass}(N_\mathfrak{p}) = \emptyset$ implies $\mathrm{Ass}(N'_\mathfrak{p}) = \emptyset$. Hence the assumptions of the lemma hold for $M \to N'$. In order to prove that φ is an isomorphism, it suffices to prove the same thing for every such $N' \subset N$. Thus we may assume N is a finite R-module. In this case, $\mathfrak{p} \notin \mathrm{Ass}(N) \Rightarrow \mathrm{depth}(N_\mathfrak{p}) \geq 1$, see Algebra, Lemma 00LL. Consider the short exact sequence

$$0 \to M \to N \to Q \to 0$$

defining Q. Looking at the conditions we see that either $Q_\mathfrak{p} = 0$ in case (1) or $\mathrm{depth}(Q_\mathfrak{p}) \geq 1$ in case (2) by Algebra, Lemma 00LX. This implies that Q does not have any associated primes, hence $Q = 0$ by Algebra, Lemma 0587. □